

Statistical and Machine Learning

Regularisation and Bayesian Methods

Instructions

All questions should first be discussed in pairs or small groups. Afterwards, we shall discuss them together. Solutions will be distributed after each session.

Exercise 1 – From OLS to Regularisation

The penalised least squares (PLS) criterion is given by

$$PLS(\beta) = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) + \lambda \text{pen}(\beta)$$

- a) Explain why OLS can fail when predictors are highly collinear or when $p > n$. What goes wrong algebraically?
- b) The PLS problem can equivalently be written as a constrained optimisation:

$$\hat{\beta}_{PLS} = \arg \min_{\beta} (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) \quad \text{subject to} \quad \text{pen}(\beta) \leq t$$

Explain in words what role the constraint $\text{pen}(\beta) \leq t$ plays. What happens to $\hat{\beta}_{PLS}$ as $t \rightarrow \infty$?

- c) The tuning parameter λ controls the strength of the penalty. Describe in words what happens to $\hat{\beta}_{PLS}$ as $\lambda \rightarrow 0$ and as $\lambda \rightarrow \infty$.

Exercise 2 – Ridge vs. Lasso

Ridge regression uses $\text{pen}(\beta) = \|\beta\|_2^2 = \sum_{k=1}^p \beta_k^2$ and Lasso uses $\text{pen}(\beta) = \|\beta\|_1 = \sum_{k=1}^p |\beta_k|$.

- a) For orthonormal covariates ($\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$), the ridge and Lasso estimators for each coefficient j reduce to simple expressions. Fill in the table below and comment on what each operation does to $\hat{\beta}_{j,LS}$.

Method	$\hat{\beta}_j(\lambda)$	Operation on $\hat{\beta}_{j,LS}$
Ridge		
Lasso		

- b) The Lasso objective is not differentiable everywhere. At which points does differentiability fail, and what does this imply for how the solution must be found?
- c) Ridge regression has a closed-form solution whereas Lasso does not. State the closed-form ridge estimator. Why does adding $\lambda \mathbf{I}_p$ to $\mathbf{X}^\top \mathbf{X}$ solve the invertibility problem that arises in OLS when $p > n$?
- d) Consider the constraint-region picture (diamond for Lasso, circle for Ridge). What do the contours refer to? What do the shaded areas refer to? Explain geometrically why Lasso tends to produce exactly zero coefficients while Ridge does not. Explain terms *shrinkage* and *sparsity* in the context of ridge regression and LASSO, respectively.

Exercise 3 – Bias, Variance and the Choice of λ

- a) Under standard linear model assumptions, OLS is unbiased: $\mathbb{E}(\hat{\beta}_{LS}) = \beta$. Show that the ridge estimator is biased by writing down $\mathbb{E}(\hat{\beta}_{PLS})$ and explaining why it differs from β .
- b) It can be shown that $Var(\hat{\beta}_{PLS}) \leq Var(\hat{\beta}_{LS})$ in the matrix sense. Explain intuitively why introducing the penalty reduces variance, and what is the price paid for this reduction.
- c) Recall that $MSE = Bias^2 + Var$. Sketch (or describe) how the bias and variance of $\hat{\beta}_{PLS}$ each change as λ increases from 0 to ∞ . What shape does the MSE curve take, and what does this imply for choosing λ ?
- d) Cross-validation is the standard method for selecting λ . Briefly describe the K -fold CV procedure applied to ridge regression. What quantity is minimised, and over what?

Exercise 4 – Extensions of Regularisation: Elastic Net and Group Lasso

- a) Suppose two predictors are perfectly correlated: $X_1 = X_2$. Describe what the Ridge and Lasso estimators each do in this case. Which method behaves more naturally, and why?
- b) The Elastic Net combines ℓ_1 and ℓ_2 penalties with a mixing parameter $\alpha \in [0, 1]$:

$$pen(\beta) = \sum_{j=1}^p (\alpha |\beta_j| + (1 - \alpha) \beta_j^2)$$

What does the Elastic Net reduce to when $\alpha = 1$? When $\alpha = 0$? Explain what intermediate values of α achieve that neither Lasso nor Ridge alone can.

- c) The Group Lasso penalty is $\sum_{g=1}^G \|\beta_{I_g}\|_2$. Explain how this differs from standard ℓ_2 regularisation, and give a concrete example of when it would be more appropriate than the ordinary Lasso.

Exercise 5 – Bayesian Inference and Bayes' Formula

- a) In the frequentist framework, μ and σ^2 in $X \sim N(\mu, \sigma^2)$ are treated as fixed unknown parameters. How does the Bayesian framework differ in its treatment of these quantities? What does Bayes' theorem then allow us to compute?
- b) Write down Bayes' theorem in the form used for Bayesian inference and name each term. Why is the normalising constant $p(\mathbf{y})$ often omitted?
- c) Interpret the Boy Who Cried Wolf update in terms of Bayes' theorem. What does it illustrate about Bayesian learning?
- d) What is a conjugate prior? State one conjugate prior pair and explain the computational advantage.

Exercise 6 – Bayesian Interpretation of Regularisation

- a) Show that the posterior mean under the normal prior coincides with the Ridge estimator and identify λ .
- b) Show algebraically that maximising the posterior is equivalent to minimising the Ridge criterion.

Exercise 7 – Conjugate priors

- a) Assume $Y \mid \theta \sim \text{Binomial}(n, \theta)$, $0 < \theta < 1$, with $p(y \mid \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}$, $y = 0, 1, \dots, n$. Let the prior distribution be $\theta \sim \text{Beta}(\alpha, \beta)$, $\alpha > 0$, $\beta > 0$ with density $p(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$, $0 < \theta < 1$. Using the Bayes theorem, prove that the beta distribution is a conjugate prior for the binomial likelihood, i.e., that a beta prior combined with a binomial likelihood produces a beta posterior.
- b) Assume $Y \mid \lambda \sim \text{Poisson}(\lambda)$, $\lambda > 0$. The probability mass function is $p(y \mid \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}$, $y = 0, 1, 2, \dots$. Let the prior distribution be $\lambda \sim \text{Gamma}(\alpha, \beta)$, $\alpha > 0$, $\beta > 0$, where α is the shape parameter and β is the rate parameter, with density $p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}$, $\lambda > 0$. Using the Bayes theorem, prove that the gamma distribution is a conjugate prior for the Poisson likelihood, i.e., that a gamma prior combined with a Poisson likelihood produces a gamma posterior.